

MOBAN Chapter 8 — Wireless Ad Hoc Networks

Comprehensive Study Guide

PART 1: THEORY SUMMARY

1. Introduction

1.1 Conventional vs. Ad Hoc Wireless Networks

Feature	Conventional WLAN (Infrastructure)	Ad Hoc / MANET
Infrastructure	Access point required	None
Topology	One-hop star	Multi-hop mesh
Administration	Centralized (AP)	Distributed
Coverage	Limited by AP range	Extended by multi-hop
Routing	Handled by AP/router	Handled by nodes themselves

IEEE 802.11 infrastructure mode: All traffic passes through an access point. Nodes communicate in one hop. The AP handles all administration.

IEEE 802.11 ad-hoc mode: Devices in range communicate peer-to-peer without infrastructure. Nodes outside direct range are reachable via multi-hop.

1.2 What is a MANET?

Mobile Ad hoc NETWORK (MANET): *A transitory association of mobile nodes that communicate with each other over wireless links and that do not depend upon any fixed support infrastructure (self-creating, self-organizing, self-administering).*

Key defining properties:

- **Self-creating:** no pre-configuration needed to join
- **Self-organizing:** network organizes itself dynamically
- **Self-administering:** no central controller

Ad hoc (Latin): *fashioned from whatever is immediately available; formed or used for specific or immediate problems.*

1.3 MANET vs. Mesh Network

	MANET	Wireless Mesh
Node roles	All nodes equal	Dedicated mesh routers + client nodes
Configuration	Spontaneous, unplanned	More planned
Mobility	All nodes mobile	Mesh routers often fixed
Scope	Broader term	Subtype of ad hoc

A wireless mesh network is a wireless multi-hop ad hoc network subtype with a more planned configuration and specific roles for mesh routers.

1.4 Why MANETs?

Four major use-case motivations:

1. **No infrastructure available** — disaster recovery, search and rescue
2. **Infrastructure too time-consuming to set up** — fast/temporary deployments (military operations, event networking)
3. **Infrastructure too expensive** — rural/remote areas, developing regions
4. **Self-organizing need** — spontaneous collaboration between devices

1.5 Origin and History

- **1970s:** Battlefield communication, US Army (origin of the concept)
- **Mid-1990s:** Commercial radios → academic research community
- **1997:** IETF MANET working group established
- **2000s onward:** Commercial products; mesh networks for internet access, IoT, etc.

1.6 Commercial Use Cases

Commercial deployments tend to be **mesh networks** (not pure device-only MANETs) because:

- Pure MANETs require all end devices to cooperate spontaneously — incentive is unclear
- Mesh networks with dedicated routers are easier to control
- Exception: military, emergency services where cooperation is enforced

Requirements driving commercial interest:

- Fast deployment
- Self-organization and self-maintenance
- Limited configuration effort

2. Characteristics of Ad Hoc Networks

2.1 Multi-hop Routing

Nodes act as **routers**, relaying traffic for others. Enables connections between nodes outside each other's direct range.

2.2 Dynamic Network Topology

Nodes move freely → variable link capacities and frequent link breaks. Routing must continuously adapt.

2.3 Autonomous Operation

No fixed infrastructure (no DHCP server, DNS, Mobile IP anchor). All functionality must operate in a **distributed manner**.

2.4 Asymmetric Links

- Different antenna characteristics, transmit power, and propagation conditions
- A node A may hear B but B may not hear A
- **Unidirectional links are possible** — routing protocols must handle or avoid them

2.5 Redundant Links

- No control over redundancy
- Causes: large computational overhead for routing updates, increased congestion, broadcasting problems

2.6 Bandwidth-Constrained, Variable Capacity Links

- Shared wireless medium → interference, fading, collisions
- Capacity varies over time and depends on neighboring transmissions

2.7 Limited Physical Security

- Open wireless medium: eavesdropping, traffic injection, denial of service
- Harder to secure than wired networks

2.8 Heterogeneous Devices

- Nodes may have different hardware (CPU, memory, battery)
- Some may be battery-powered → energy is a design constraint

2.9 Network Splitting and Merging

- Mobile nodes may cause a network to partition or two separate networks to merge
- Protocols must handle these events gracefully

3. Impact on Protocol Design

The challenging environment of MANETs affects **every layer** of the protocol stack:

Challenge	Protocol Area
Fair channel access, hidden/exposed terminals	MAC protocols (see Ch. 2)

Challenge	Protocol Area
Establishing/maintaining routes between any two nodes	Routing protocols
Efficiently propagating control info to all nodes	Broadcasting techniques
Finding services in a dynamic network	Resource & service discovery
Assigning unique IP addresses without DHCP	Address assignment
TCP assumes fixed, reliable links — not valid here	Transport protocols
Accessing the Internet from an ad hoc network	Internet connectivity
Detecting/preventing attacks	Security
Preventing nodes from refusing to forward traffic	Cooperation enforcement

4. Routing in Ad Hoc Networks

4.1 Why Traditional Routing Fails

Standard fixed-network routing protocols (e.g., RIP, OSPF) are designed for:

- Slow topology changes → infrequent updates are sufficient
- High bandwidth for control traffic → large routing tables acceptable

In MANETs these assumptions break:

- Topology changes fast (mobility) → slow convergence = stale routes
- Bandwidth is scarce → large routing overhead is unacceptable

4.2 Routing Challenges

Finding a route from A to B is hard when:

- Links are asymmetric
- Bandwidth for control messages is limited
- Control message propagation is unreliable (wireless broadcast)
- The environment is highly dynamic
- Link capacity varies and link breaks occur frequently
- Computation power is low
- Energy consumption matters
- Security must be guaranteed

4.3 L2 vs. L3 Routing

	Mesh Under (L2)	Route Over (L3)
Forwarding based on	MAC (Layer 2) addresses	IP (Layer 3) addresses
IP routing inside mesh	No — single broadcast domain	Yes — collection of link-local scopes

	Mesh Under (L2)	Route Over (L3)
Example	IEEE 802.11s (HWMP)	OLSR, AODV

5. Classification of Routing Protocols

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Ad hoc routing protocols
├── Proactive (table-driven)
│   ├── Distance vector: DSDV, WRP
│   └── Link state: OLSR, OLSRv2
├── Reactive (on-demand)
│   ├── Source routing: DSR
│   └── Hop-by-hop: AODV, DYMO
├── Hybrid
│   └── ZRP (Zone Routing Protocol)
└── Location-based
    └── LAR, DREAM

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6. Proactive Routing Protocols

Definition: Maintain up-to-date routes to every possible destination at all times.

How: Exchange routing control information throughout the entire network periodically and upon link changes.

Advantage: Routes are always ready — no delay when traffic needs to start.

Disadvantage: Continuous overhead even when no data is being sent; poor scalability.

6.1 Distance Vector Routing (Background)

Based on the **Distributed Bellman-Ford (DBF)** algorithm (same as RIP for fixed networks).

Each node maintains:

- **Distance table:** distance to each destination via each neighbor
- **Routing table:** best next hop to each destination

Nodes periodically exchange **distance vectors** = (destination, distance) with neighbors.

Problems in MANETs:

- **Counting to infinity:** When a route breaks, nodes may keep incrementing the metric forever until they agree the route is gone
- **Routing loops:** Temporary inconsistencies cause packets to loop

Routing anomalies are significantly worse in MANETs due to faster topology changes!

6.2 DSDV — Destination Sequenced Distance Vector

Type: Proactive, distance vector

Key innovation: Adds **sequence numbers** to routing updates to guarantee loop-free routing.

- Distance vector format: **(destination, metric, sequence_number)**
- Sequence number is maintained by the **destination** node itself
- Ensures all updates are applied in order → no loops

How DSDV avoids loops:

- When a route breaks: destination increments its sequence number and advertises **(dest, ∞, seq_odd)**
- Intermediate nodes receiving a higher sequence number immediately discard lower-sequence-number routes
- Old stale routes with lower sequence numbers are ignored, preventing loops

Two types of updates:

Update Type	Content	Frequency
Full dump	All routing info	Infrequent
Incremental	Only changes since last full dump	Frequent

Damping: Delays advertisement of unstable routes. When a route changes rapidly, DSDV waits before advertising it (stores time between first and best announcement of a path; inhibits update if route seems unstable).

Drawbacks:

- Trade-off between overhead and convergence latency
- Parameters (periodic update interval, settling time) must be tuned

6.3 WRP — Wireless Routing Protocol

Type: Proactive, distance vector

Key features:

- Update messages include **(destination, distance, predecessor)** — the predecessor info helps avoid loops
- Sends routing updates **periodically** and **upon topological changes**
- All nodes maintain hello messages to detect neighbors

WRP overhead: Independent of traffic load — each node sends a hello message every 10 seconds regardless of data traffic. For N nodes: **N/10** packets/second.

6.4 OLSR — Optimized Link State Routing Protocol

Type: Proactive, link state

Two key optimizations over standard link state flooding:

1. Optimized flooding using Multipoint Relays (MPR)

2. Reduced link state info: Only MPR nodes generate and forward link state information

Multipoint Relays (MPR)

Goal: Reduce the number of retransmissions during flooding.

Concept: Each node selects a subset of its 1-hop neighbors (the MPR set) such that:

- Every **2-hop neighbor** is reachable through at least one MPR
- The MPR set is **minimal** (fewest nodes possible)

Flooding rule:

- A node only rebroadcasts a flooding packet if it received it from one of **its MPRs**
- This dramatically reduces duplicate transmissions

MPR Set Construction Algorithm:

1. Identify all **2-hop neighbors** of the node
2. Find **forced MPRs**: 1-hop neighbors that are the only path to reach a specific 2-hop neighbor
3. Add those forced MPRs to the MPR set
4. Check which 2-hop neighbors are still uncovered
5. Greedily select the 1-hop neighbor covering the most uncovered 2-hop neighbors
6. Repeat until all 2-hop neighbors are covered

Example:

```
Node 1's 1-hop neighbors: {3, 7, 9, 11, 12}
Node 1's 2-hop neighbors: {6, 8, 2, 10, 14, 16, 13, 5}

Forced MPRs:
Node 12 → only one that reaches {5, 13, 16}
Node 11 → only one that reaches {10, 14}
Node 7 → only one that reaches {6, 8}

Remaining uncovered 2-hop NB: {2}
Reachable via node 3 or node 9

Possible MPR sets: {12, 11, 7, 3} or {12, 11, 7, 9}
```

Link state reduction: Only MPR nodes broadcast their link state information (Topology Control messages). Non-MPR nodes do not generate TC messages. This significantly reduces the volume of link state info flooding the network.

OLSRv2 (Updated Version)

- Same core algorithms as OLSR
- More flexibility, simplicity, efficiency, modularity
- Supports **link metrics other than hop count**

- Uses stand-alone components: MANET Neighborhood Discovery Protocol (NHDP), generalized TLV-based packet format

7. Reactive Routing Protocols

Definition: Routes are set up **only when needed** and maintained **only as long as needed**.

How:

1. **Route discovery:** Broadcast Route Requests (RREQ) — wait for Route Reply (RREP)
2. **Route maintenance:** Keep active routes alive; send Route Errors (RERR) on link breaks

Advantage: No overhead for routes that are never used.

Disadvantage: Latency before first packet can be sent (route discovery delay).

7.1 AODV — Ad hoc On-demand Distance Vector

Type: Reactive, hop-by-hop forwarding

Three message types:

Message	Direction	Purpose
RREQ (Route Request)	Flooded outward	Source requests route to destination
RREP (Route Reply)	Unicast back to source	Destination (or intermediate node with fresh route) responds
RERR (Route Error)	Broadcast upstream	Notifies sources of a broken link

RREQ Propagation

1. Source broadcasts RREQ with a unique ID
2. Every intermediate node rebroadcasts the RREQ **exactly once** (identified by source IP + RREQ ID)
3. RREQ records the reverse path implicitly at each node
4. Destination receives RREQ → sends RREP back along reverse path

RREP Propagation

1. Destination (or intermediate node with a valid cached route) sends RREP
2. Each node forwarding RREP records the **forward path** entry
3. Source receives RREP → data flow can begin

Routing Tables in AODV

- Only **active routes** are kept in routing tables
- Unused entries **time out** automatically
- Routing table entry: (destination, next-hop, sequence-number, lifetime)

Route Error (RERR)

1. Node detects a link break (neighbor gone silent)
2. Broadcasts RERR to all sources using that link
3. Sources initiate a new route discovery

Control Packet Count for AODV

- **RREQ**: Broadcast by every node except the destination → **(N-1) RREQs** for N nodes
- **RREP**: Unicast, forwarded by every node on the path from destination to source → **h RREPs** for a path of h hops

Total overhead per route discovery: $(N-1) + h$ control packets

AODV Expanding Ring Search

Optimization to reduce overhead when destination is nearby:

1. **Phase 1**: Send RREQ with **TTL = 2** (reaches only 1-hop and 2-hop neighbors)
 - If destination found within 2 hops → RREP received, done
 - Control overhead: $1 + (\text{number of 1-hop neighbors})$ RREQs
2. **Phase 2**: If no RREP received, send RREQ with **TTL > max hops** (floods entire network)

When expanding ring search helps: When many destinations are close (within TTL). For distant destinations, it actually adds overhead (two phases instead of one).

7.2 DSR — Dynamic Source Routing

Type: Reactive, source routing

Key difference from AODV: The **full route is stored in every data packet** (source routing). Intermediate nodes don't maintain routing tables.

Route Discovery

1. Source broadcasts RREQ with a **Route Record (RR)** — initially empty
2. Each intermediate node appends its own address to the RR before rebroadcasting
3. RREQ propagation stops when:
 - Node already received this RREQ (discard)
 - Node's address already in the RR (loop detected)
4. Destination receives RREQ → sends RREP with the complete route

Route Reply

- Destination sends RREP containing the full route from RREQ's RR
- RREP is forwarded along the **reverse route** (if links are bidirectional)
- If links are unidirectional: piggyback RREP in a new RREQ in the reverse direction

Packet Forwarding

- Each data packet carries the full list of intermediate nodes
- Intermediate nodes simply read the next hop from the source route in the packet

DSR Optimizations

- **RREQ hop limits:** Limit RREQ propagation (like AODV's TTL)
- **Route caching:** Nodes cache routes overheard from passing RREQ/RREP packets → reduces discovery overhead
- **Delay RREP:** Prevents RREP storms when multiple nodes respond simultaneously

7.3 DYMO — Dynamic MANET On-demand Routing

- Successor of AODV (also known as AODVv2)
- Same algorithms as AODV, slightly simplified implementation
- **Path accumulation feature:** Nodes append their info to RREQ/RREP, giving better topology knowledge
- Uses a generalized packet/message format (compatible with OLSR message format)

8. Hybrid Routing: ZRP

ZRP — Zone Routing Protocol

Concept: Combine proactive and reactive approaches using a **zone radius** parameter (e.g., $r = 2$ hops).

Inside the zone ($\leq r$ hops)	Outside the zone ($> r$ hops)
Proactive intrazone routing	Reactive interzone routing
Routes maintained at all times	Routes established on demand
Low latency for nearby nodes	Low overhead for distant nodes

Interzone routing uses **bordercasting**: send route requests to border nodes of the zone, which relay them into adjacent zones.

Trade-off: Zone radius r must be tuned. Larger r → more proactive overhead; smaller r → more reactive latency.

9. Hierarchical Routing: Clustering

Goal: Improve scalability by introducing hierarchy.

How:

- Network is partitioned into **non-overlapping clusters**
- Each cluster has one **cluster head**
- Cluster heads take a leading role in disseminating routing information
- **Gateway nodes** connect adjacent clusters

Example: RREQs are only forwarded by cluster heads via gateway nodes — dramatically reduces flooding scope.

10. Location-Based Routing

Concept: Use physical location information to **limit flooding** of control messages.

How it works:

1. A **location information service** provides nodes with the position of the destination
2. Control packets (or data packets) are forwarded only toward the destination's direction

Three types of location services:

Type	How
Proactive location database	Specific nodes act as databases; nodes register/query position
Proactive location dissemination	All nodes periodically broadcast their current location
Reactive location systems	Position is queried on-demand (like reactive routing)

10.1 DREAM — Distance Routing Effect Algorithm for Mobility

- Nodes periodically broadcast their physical location
- **Distance effect:** Far-away nodes move at lower apparent angular speed → need fewer updates (packets have a lifetime proportional to distance)
- **Speed effect:** Faster-moving nodes broadcast more frequently
- Data packets are **partially flooded** toward the cone in the direction of the destination

10.2 LAR — Location Aided Routing

Reactive protocol that uses location information to limit the RREQ flood area.

Two schemes:

Scheme 1 — Expected Zone:

- Source calculates an **expected zone** for destination D based on last known position P_0 at time t_0 and estimated speed
- Expected zone is a circle of radius $r = (t_1 - t_0) \times \text{speed_estimate}$ around P_0
- RREQ is only rebroadcast by nodes within a **request zone** that encloses the expected zone

Scheme 2 — Distance-Based:

- Source calculates its distance to the destination and puts it in the RREQ
 - Only nodes that are **closer to the destination** than the source relay the RREQ
 - Forward condition: $\text{DSTD}(\text{relay}) < \text{DSTD}(\text{source}) + \Delta$
-

11. Routing Metrics

11.1 Metrics to Minimize (lower = better)

Metric	Description
Hop count (h)	Number of hops; simplest metric. Favors long-range but often less stable links
ETX — Expected Transmission Count	Expected number of transmissions needed for a packet to be received error-free at the destination. Calculated from forward and reverse delivery ratios. Range: $[1, \infty)$
Loss rate	Fraction of packets lost on a link
Delay	Transmission + propagation + queuing delay
Congestion	Number of neighbors that can overhear a transmission (interference proxy)

11.2 Metrics to Maximize (higher = better)

Metric	Description
Bandwidth	Available throughput on the path (bottleneck link determines path bandwidth)
Stability	Connectivity lifetime of links
Battery runtime	Remaining energy at forwarding nodes

11.3 ETX Detail

ETX for a link = $1 / (\text{forward_delivery_ratio} \times \text{reverse_delivery_ratio})$

ETX for a path = sum of ETX values for each link on the path

Example:

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Link predicted throughput: 100%  50%  50%  33%  20%
ETX for link:             1       2    2     3    5
Path ETX:                 1 + 2 + 2 + 3 + 5 = 13

```

A path with 5 long-range unreliable links can have a higher ETX than a path with more shorter, reliable hops.

11.4 Combined Cost Function

$$\text{Cost} = \alpha \cdot h + \beta \cdot i + \gamma \cdot d + \delta \cdot r$$

where:

- h = hop count
- i = interference

- d = delay
- r = some reliability metric
- $\alpha, \beta, \gamma, \delta$ = weights

11.5 Performance Metrics for Comparing Protocols

- **Packet delivery ratio** (PDR)
- **Routing overhead** (control packets / data packets)
- **End-to-end delay**
- **Throughput**

Key insight: Every MANET protocol can outperform the others in a specific networking context. Performance depends heavily on: network size, density, traffic, mobility, link asymmetry, etc. There is no universally best protocol.

12. Broadcasting in MANETs

12.1 Why Broadcasting is Needed

Broadcasting is used for:

- RREQ propagation (reactive protocols)
- Link state flooding (proactive protocols)
- Hello messages (neighbor discovery)

Naive approach: **Blind flooding** — every node rebroadcasts every packet it receives once.

12.2 The Broadcast Storm Problem

Problem: In dense networks, blind flooding causes massive redundancy:

- When A broadcasts, both B and C receive it
- B and C both rebroadcast at nearly the same time → **collision**
- Node D receives the same packet multiple times (from B and from C) — only one copy was needed
- In a large network: exponential explosion of rebroadcasts

Consequences:

- Network cannot transport normal traffic
- Severe medium contention and collision
- This is called the **broadcast storm problem**

12.3 Broadcast Solutions

Technique	How
Random jitter	Add a random delay before rebroadcasting → desynchronizes transmissions, reduces simultaneous collisions

Technique	How
Counter-based	Node only rebroadcasts if it has received fewer than k copies (rebroadcasting becomes less likely as copies accumulate)
Distance-based	Only rebroadcast if the sender is far away (large coverage gain)
Location-based	Only rebroadcast if the additional coverage area is large
2-hop neighbor-based (MPR)	Only designated MPR nodes rebroadcast — the OLSR approach

The **MPR approach** is the most principled solution: it guarantees that all 2-hop neighbors are covered while minimizing retransmissions.

13. Underlying Technologies

13.1 IEEE 802.11s (WLAN Mesh Standard)

- Integrated mesh networking solution extending the **IEEE 802.11 MAC** standard
- Supports unicast, multicast, and broadcast over **self-configuring multi-hop topologies**
- **Layer 2 mesh:** the mesh appears as a single broadcast domain to higher layers
- Runs on top of 802.11a/b/g/n physical layers
- **Default routing protocol:** HWMP (Hybrid Wireless Mesh Protocol) — based on AODV combined with tree-based proactive routing
- Other routing protocols can optionally be supported
- Used in: One Laptop per Child (OLPC) project, open80211s (Linux kernel)

13.2 IEEE 802.15.4 Multi-hop Networks (IoT)

- Used for low-power, low-rate IoT/sensor networks
- **Routing mechanism:** RPL — establishes a **DODAG** (Destination Oriented Directed Acyclic Graph)
- DODAG provides a path from every node to a root router (**MP2P traffic = upward traffic**)
- Nodes have a **parent-child relationship** in the DAG
- Multiple parents provide redundancy
- P2P traffic (between two sensors) is supported but more complex

13.3 BLE Meshing

BLE (Bluetooth Low Energy) supports two communication modes:

Mode	Description
Broadcaster-Observer	No connections; uses 3 advertising channels; one device broadcasts, others observe
Central-Peripheral	Connection-based; hops over 37 data channels

Two BLE meshing flavors:

Rebroadcasting mesh (connection-less):

- Flood all messages to all nodes via broadcasts
- Need to control flooding (broadcast storm problem)
- Example: Nordic nRF OpenMesh

Spanning tree mesh (connection-based):

- Broadcasting used in startup phase to discover neighbors
- Constructs a spanning tree; each node has up to 7 connections (6 as central, 1 as peripheral)
- Example: Fruitymesh

14. Summary: Protocol Comparison

Protocol	Type	Subtype	Key Idea	Overhead	Route Latency
DSDV	Proactive	Distance vector	Sequence numbers for loop-free routing	High (always)	Zero
WRP	Proactive	Distance vector	Predecessor info to avoid loops	High (always)	Zero
OLSR	Proactive	Link state	MPR reduces flooding overhead	Medium-High	Zero
AODV	Reactive	Hop-by-hop	RREQ flood + RREP unicast	Low when idle	On first packet
DSR	Reactive	Source routing	Route stored in packet header	Low when idle	On first packet
DYMO	Reactive	Hop-by-hop	AODV successor with path accumulation	Low when idle	On first packet
ZRP	Hybrid	—	Proactive inside zone, reactive outside	Tunable	Low nearby
LAR	Location-based	Reactive	Limit RREQ flood to expected zone	Low	On first packet
DREAM	Location-based	Proactive	Directional partial flood	Medium	Zero

When to prefer proactive (e.g., WRP/OLSR):

- High connection frequency
- Nodes communicate often with many destinations
- Delay budget is tight

When to prefer reactive (e.g., AODV):

- Sparse traffic (few connections set up per unit time)

- Destination may not be needed → avoid maintaining unnecessary routes
- Network is large but sparsely used

PART 2: EXERCISES AND SOLUTIONS

Exercise 1: AODV / OLSR / WRP — 16-Node Network

Network Setup

A 16-node wireless ad hoc network with link speeds given in Mbps.

Link speed to delay conversion:

Link speed (Mbps)	Data packet (1000 bytes = 8000 bits)	Control packet (500 bytes = 4000 bits)
1	8 ms	4 ms
2	4 ms	2 ms
4	2 ms	1 ms
8	1 ms	0.5 ms

Assumptions:

- Propagation and processing delays = 0
- All links bidirectional (symmetric)
- No transmission errors
- Data packets: 1000 bytes; Control packets: 500 bytes

Part I: AODV — Node 6 → Node 16, Metric = Delay

Key AODV principle: The RREQ that **arrives first** at the destination determines the path (metric = delay).

Q1: What is the resulting path?

Convert all link speeds to per-packet delays, then find the path from 6 to 16 with **minimum total control-packet delay** (RREQ uses control-packet size).

Resulting path: 6 → 3 → 9 → 1 → 11 → 14 → 15 → 16

Control packet delays along this path:

- 6 → 3: 2 ms (4 Mbps link)
- 3 → 9: 0.5 ms (8 Mbps link)
- 9 → 1: 1 ms (4 Mbps link)
- 1 → 11: 1 ms (4 Mbps link)
- 11 → 14: 1 ms (4 Mbps link)
- 14 → 15: 1 ms (4 Mbps link)
- 15 → 16: 0.5 ms (8 Mbps link)

- **Total RREQ travel time: 7 ms**

Q2: How long does it take to set up the connection?

- RREQ arrives at node 16 after **7 ms**
- RREP travels back along the same path, also **7 ms**
- **Total setup time: 14 ms**

Q3: How many control packets are needed?

- **RREQ:** Broadcast by every node except the destination → **15 RREQs** (16 nodes – 1)
- **RREP:** Unicast, forwarded by every node on the path from 16 back to 6
 - Path = 6 → 3 → 9 → 1 → 11 → 14 → 15 → 16 → nodes 16, 15, 14, 11, 1, 9, 3 forward the RREP → **7 RREPs**
- **Total: 22 control packets**

Q4: Routing tables of nodes 1 and 12 after a long time?

- **Node 12:** Not on the path → routing table remains **empty** (AODV only keeps entries for active routes)
- **Node 1:** On the path → two entries:
 - Destination 6, next hop = 9
 - Destination 16, next hop = 11

Q5: Path if all links had the same delay?

When all link delays are equal, the RREQ that arrives first is the one that traversed the **fewest hops**.

There are **4 paths of 4 hops** between node 6 and node 16. All arrive simultaneously. The actual path chosen would depend on processing delays and MAC behavior (non-deterministic in this idealized model).

Part II: OLSR — Node 6 → Node 16

Q1: Path with metric = hop count?

All 4 shortest paths have **4 hops**. Any of them is valid. Example: **6 → 7 → 1 → 12 → 16**

Metric = 4 hops

Q2: Path with metric = bandwidth?

Bandwidth of a path = **minimum link bandwidth** along the path (bottleneck).

Goal: **maximize the minimum link bandwidth**.

Best path: **6 → 7 → 8 → 5 → 12 → 1 → 11 → 14 → 15 → 16**

- All links on this path have bandwidth ≥ 4 Mbps
- **Path metric (bandwidth) = 4 Mbps**

Note: This path has more hops (9 hops) than the hop-count-optimal path (4 hops), but achieves higher bandwidth.

Q3: MPR set of node 1

Step-by-step MPR construction for node 1:

1. **1-hop neighbors of 1:** {3, 7, 9, 11, 12}
2. **2-hop neighbors of 1:** {6, 8, 2, 10, 14, 16, 13, 5}
3. **Forced MPRs** (only neighbor that can reach a given 2-hop neighbor):
 - 2-hop NB {5, 13, 16}: only reachable via **12** → 12 is a forced MPR (covers {5, 13, 16})
 - 2-hop NB {10, 14}: only reachable via **11** → 11 is a forced MPR (covers {10, 14})
 - 2-hop NB {6, 8}: only reachable via **7** → 7 is a forced MPR (covers {6, 8})
4. **Remaining uncovered 2-hop NB:** {2}
 - Reachable via **3** or **9** (either works)
5. **Possible minimal MPR sets:**
 - {12, 11, 7, 3} or {12, 11, 7, 9}

Part III: AODV vs. WRP Overhead (f connections/second)

Setup:

- Node 1 = server
- Nodes 2, 8, 12 each initiate f connections/second to node 1
- AODV: each connection requires a full RREQ/RREP cycle; metric = delay
- WRP: converged; only hello messages every 10 seconds per node; 16 nodes total

Optimal paths (metric = delay) for each source:

Source	Path to node 1	Hops
Node 2	2 → 9 → 1	2
Node 8	8 → 5 → 12 → 1	3
Node 12	12 → 1	1

AODV overhead:

Per connection per source (RREQ broadcasts entire network = 15 RREQs):

Source	RREQs	RREPs	Total/connection
Node 2	15	2	17
Node 8	15	3	18
Node 12	15	1	16
Total per 3 connections	45	6	51

AODV overhead = 51·f packets/second

WRP overhead:

- Each of 16 nodes sends a hello every 10 seconds
- **WRP overhead = $16/10 = 1.6$ packets/second** (independent of f)

Q1: At what f do they have the same overhead?

$$51 \cdot f = 1.6$$

$$f = 1.6/51 \approx 0.0314 \text{ connections/second} \approx 1.88 \text{ connections/minute}$$

Q2: Which is best for large f ?

For large f , **WRP is better** because:

- AODV overhead grows linearly with f ($51 \cdot f$)
- WRP overhead is constant (1.6 pkt/s), independent of f

Q3: AODV with Expanding Ring Search (TTL = 2)

With TTL = 2, the RREQ only reaches 1-hop and 2-hop neighbors. Nodes 2, 8, and 12 can all reach server 1 in ≤ 2 hops, so the first phase succeeds.

Source	Phase 1 RREQs	RREPs
Node 2	3 (nodes 2, 3, 9)	2
Node 8	3 (nodes 8, 7, 5)	2
Node 12	4 (nodes 12, 5, 13, 16)	1
Total	10	5 = 15 total

Expanded ring search overhead = $15 \cdot f_{\text{exp}}$ packets/second

How many more connections for the same overhead?

$$15 \cdot f_{\text{exp}} = 51 \cdot f$$

$$f_{\text{exp}} / f = 51/15 = 3.4$$

→ **3.4× more connections** can be supported with the same control packet overhead.

Exercise 2: AODV Normal vs. Expanding Ring Search — 11-Node Network

Network Setup

- 11 nodes, 4 connections: $1 \rightarrow 11$, $4 \rightarrow 5$, $7 \rightarrow 9$, $2 \rightarrow 10$
- No packet loss, AODV routing

Question: Which technique has lower overhead — normal RREQ or expanding ring search (TTL = 2)?

Normal RREQ Propagation

- Each RREQ is rebroadcast by every node except the destination → **(N-1) = 10 RREQs per destination**
- 4 destinations → **40 RREQs total**

Expanding Ring Search (TTL = 2)

Rules:

- If destination within 2 hops: only Phase 1 (TTL=2) needed
- If destination more than 2 hops away: Phase 1 + Phase 2 (full flood)

Phase 1 RREQs = 1 (source) + (number of 1-hop neighbors of source, which rebroadcast)

Connection	Path length	Phase 1	Phase 2	Total
1 → 11	4 hops	1 + 2 = 3	10	13
4 → 5	1 hop	1 + 2 = 3	—	3
7 → 9	2 hops	1 + 2 = 3	—	3
2 → 10	3 hops	1 + 3 = 4	10	14
Total				33

Conclusion: Expanding ring search = **33 RREQs** vs. normal = **40 RREQs**.

Expanding ring search is better for this network because many routes are short (1-2 hops) and don't need the full flood.

Key insight: Expanding ring search is most beneficial when many destinations are close. If most destinations are far away, the two-phase approach actually adds overhead (you pay for both phases).

Exercise 3: OLSR MPR — 25-Node Network

Network Setup

- 25 nodes, OLSR routing
- Find a **minimal MPR set for node 1**

Solution: Step-by-Step

Node 1's 1-hop neighbors: {4, 5, 7, 8, 9, 10, 11, 13, 15, 16, 18, 22}

Node 1's 2-hop neighbors: {6, 12, 14, 17, 19, 20, 21, 23, 24, 25, ...} (*all nodes reachable in exactly 2 hops*)

Step 1 — Find forced MPRs (only 1-hop neighbor of 1 that can reach a given 2-hop neighbor):

2-hop NB forced via	Forced MPR	Covers
Node 12	Only reachable via ...	Identify from topology
Node 22	Forced MPR	From 1's neighbors

Step 2 — Identify remaining uncovered 2-hop neighbors after forced MPRs are added

Step 3 — Greedily select additional 1-hop neighbors to cover remaining 2-hop neighbors with fewest extra nodes

Possible MPR sets (8 valid solutions of size 5):

```
{4, 22, 5, 13, 18}
{4, 22, 5, 15, 20}
{4, 22, 5, 15, 18}
{4, 22, 5, 15, 16}
{4, 22, 10, 13, 18}
{4, 22, 10, 15, 20}
{4, 22, 10, 15, 18}
{4, 22, 10, 15, 16}
```

Key observations:

- Nodes 4 and 22 are **always forced MPRs** (forced because they are the only path to certain 2-hop neighbors)
- Either node 5 or 10 must be chosen (to reach 2-hop NB node 6)
- Either node 13 with 18, or node 15 with {16, 18, or 20}, must be added to cover remaining 2-hop neighbors
- **No single solution** — there are 8 valid minimal MPR sets, each of size 5 out of 12 total 1-hop neighbors

PART 3: EXAM PREPARATION — KEY QUESTIONS

The following questions are taken directly from the slides as study prompts:

Conceptual Questions

1. Why is routing in ad hoc networks different from routing in fixed networks?

- Fast topology changes → slow convergence of traditional protocols
- Scarce bandwidth → high overhead of link state flooding is unacceptable
- Unreliable broadcast medium for control messages
- Low power, asymmetric links, mobile nodes

2. Give the main classes of ad hoc routing protocols and a one-sentence description of each.

- **Proactive:** Maintains routes to all destinations at all times via continuous exchange of control info
- **Reactive:** Sets up routes on demand when communication is needed
- **Hybrid:** Uses proactive routing within a zone and reactive routing outside it
- **Location-based:** Uses physical location information to limit where control messages are propagated

3. Describe two strategies for reactive routing. Give an example for each.

- **Source routing (DSR):** The full route is stored in every packet; intermediate nodes follow the stored route without maintaining tables
- **Hop-by-hop routing (AODV):** Each node maintains its own routing table; packets carry only the destination address and each node decides the next hop

4. Describe two strategies for proactive routing. Give an example for each.

- **Distance vector (DSDV, WRP):** Nodes exchange distance vectors with neighbors; each node maintains distances to all destinations
- **Link state (OLSR):** Nodes flood link state information to build a global topology map and run Dijkstra locally

5. **What is a routing metric?** A routing metric is a value used to evaluate the quality of a path. It guides the selection of the best route. Examples: hop count, ETX, delay, bandwidth, stability.

6. Describe 2 broadcasting strategies in MANETs.

- **Random jitter:** Nodes add a random delay before rebroadcasting to desynchronize transmissions and reduce collisions
- **MPR-based:** Only multipoint relay (MPR) nodes rebroadcast, ensuring all 2-hop neighbors are covered with minimal redundancy

7. Explain: MPR, source routing

- **MPR:** A subset of a node's 1-hop neighbors chosen such that every 2-hop neighbor is covered; used in OLSR to optimize flooding
- **Source routing:** A routing approach (used in DSR) where the sender specifies the complete path in the packet header; intermediate nodes do not need routing tables

8. Give an example use case and explain why ad hoc networking/meshing is useful.

- *Disaster recovery:* After an earthquake, infrastructure is destroyed. First responders can form a MANET with their devices, enabling communication without relying on base stations or internet access
- *Military operations:* Soldiers in the field need a self-organizing network that works without fixed infrastructure and reconfigures automatically when soldiers move

Quick-Reference Formulas

Quantity	Formula
Link delay (data)	$8000 \text{ bits} / \text{link_speed_in_bps}$
Link delay (control)	$4000 \text{ bits} / \text{link_speed_in_bps}$
AODV RREQs (full flood)	$N - 1$ (N = total nodes)
AODV RREPs	h (h = hops in path)
AODV total overhead per connection	$(N-1) + h$

Quantity	Formula
WRP overhead	N / T_{hello} packets/second
ETX for a link	$1 / (p_{\text{forward}} \times p_{\text{reverse}})$
ETX for a path	$\sum \text{ETX}_{\text{link}}$ over all links
AODV overhead (f connections/s)	$(\sum \text{over all sources: } (N-1) + h_i) \times f$
Connection equivalence (ring search)	$f_{\text{normal}} \times \text{overhead}_{\text{normal}} = f_{\text{ring}} \times \text{overhead}_{\text{ring}}$

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